

Sneha Mahapatra

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EDUCATION

University of Massachusetts, Amherst

Master of Science in Computer Science

Amherst, MA

Sep 2021 – May 2023

Purdue University

Bachelor of Science in Computer Engineering

West Lafayette, IN

Aug 2017 – Dec 2020

EXPERIENCE

Microsoft

Redmond, WA

AI Software Engineer II

Dec 2025 – Present

- Architected an end-to-end agentic bug-triage system that orchestrates ML and LLM pipelines to transform raw user feedback and metadata into actionable, deduplicated engineering bugs and Root Cause Analysis reports.
- Analyzed 6–7 million feedback items monthly using actionability classification, text embeddings, spherical k-means clustering, and cohesion/semantic scoring to identify actionable issues and deduplicate related feedback.
- Designed the LLM orchestration layer to generate issue titles and descriptions, reclassify candidate issues, and route high-confidence results for automated bug creation, achieving an 75% actionability score.

AI Software Engineer

- Architected and built an MCP search and analytics platform used by 500+ daily users, orchestrating semantic retrieval, aggregations, classification, and sentiment analysis across millions of customer feedback records and 3,000+ searchable fields.
- Developed AI-powered sentiment and trend analysis adopted across 20+ Microsoft product teams, using LLMs and machine learning to transform large-scale customer feedback into concise, actionable narratives.
- Improved LLM reliability and production stability through tokenizer validation, token-budget optimization, and inference pipeline enhancements, reducing model failure rates from 33% to nearly 0%.

Superluminal

Software Developer

May 2023 – Present

- Developed desktop version of client medical web application, broadening user accessibility by 100%.
- Architected and prototyped a screen-recording feature for demos, reducing debugging time by 60%.
- Optimized backend services and controllers integrated into APIs supporting over 10,000 clients.
- Increased triage efficiency by 350% through Azure-compatible logging integrations.
- Skills Acquired: C#, .Net, SQL, SQLite, Azure.

BioNLP

Application Developer Researcher

Sep 2021 – May 2023

- Led development of an iOS application to help low-income users access government benefits.
- Integrated Mapbox and Firebase to provide location-based resources across 45 states.
- Fine-tuned a BERT NLP model to generate tailored search results from web-scraped data.
- Built crawlers aggregating over 1,000 Google search results into a structured database.
- Skills Acquired: iOS mobile development, Firebase, Mapbox, Natural Language Processing.

PROJECTS

Senior Design Project

Machine Learning Engineer & Software Developer

Jan 2020 – Dec 2020

- Architected a real-time ASL translator using computer vision with up to 94% accuracy for static gestures and 92% for movement gestures.
- Developed an iOS application integrating Google Mediapipe with a custom ASL model with 9 user-selectable categories.
- Led the senior design team through a year-long project lifecycle.
- Skills Acquired: Computer Vision Model Development, iOS Development, Integrating CV ML Model, Project Management.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C#, MATLAB, C++
- **Web/Mobile:** JavaScript, React Native, Node.js
- **ML/AI:** PyTorch, NumPy, Pandas, Tableau
- **Cloud/Tools:** Azure, SQL, Git, Windows, Jira